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Meta-analysis

The prevalence of schizophrenia in mainland China: evidence from epidemiological surveys

Long J, Huang G, Liang W, Liang B, Chen Q, Xie J, Jiang J, Su L. The prevalence of schizophrenia in mainland China: evidence from epidemiological surveys

Objective: Schizophrenia is a severe mental disorder. Its prevalence appears inconsistent in different regions of China; thus, we conducted this meta-analysis to estimate the prevalence of schizophrenia in mainland China.
Method: Studies on the prevalence of schizophrenia in mainland China were identified from electronic databases up to July 2013. Meta-analysis was used for prevalence of schizophrenia estimate.
Results: For lifetime prevalence of schizophrenia, prevalence of 5.44 per 1000 (overall), 5.33 per 1000 (males), 5.51 per 1000 (females), 6.60 per 1000 (urban) and 4.73 per 1000 (rural) were estimated; there was no significant difference between males and females, while prevalence for urban dwellers was higher than for rural (OR = 1.44, 95%CI: 1.30–1.59). For point prevalence of schizophrenia, prevalence of 4.62 per 1000 (overall), 4.63 per 1000 (males), 4.95 per 1000 (females), 5.15 per 1000 (urban) and 4.44 per 1000 (rural) were estimated; no statistical difference was found in males and females, but prevalence for urban dwellers was higher than for rural (OR = 1.20; 95%CI: 1.02–1.41).
Conclusion: The prevalence estimate results were consistent with the international prevalence estimate of schizophrenia in mainland China. There was no significant difference between males and females, while prevalence for urban dwellers was higher than for rural.

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Key words: schizophrenia; prevalence; epidemiology; China; meta-analysis

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Summations

- This meta-analysis found that mainland China has a wide range of prevalence estimates of schizophrenia. However, the pool prevalence was stable.
- Location distribution affects the prevalence estimates significantly, but not time or gender distribution.

Considerations

- This is a comprehensive meta-analysis summarizes the prevalence of schizophrenia in mainland China. However, some areas such as Chongqing, Tianjin, Shanxi Province, Nei Monggol Autonomous Region, and Ningxia Hui Autonomous Region have not been sufficiently studied.
- Due to differences in the design and quality of the studies in assorted and diverse areas, the heterogeneity cannot be avoided.

Introduction

Schizophrenia is defined as a mental illness of unknown etiology, tending toward young onset, which is often accompanied by disordered perception, thinking, emotion, behaviour and other aspects of disorders and many uncoordinated mental activities (1). According to the World Health Organization (WHO), there are about 450 million people with different degrees of mental illness worldwide, among whom about 60 million suffered from schizophrenia (2). It is reported that people with schizophrenia have a higher probability than the general population to suffer from HIV infection, osteoporosis, sexual dysfunction, obstetric complications, cardiovascular diseases, overweight and diabetes (3). Schizophrenia not only seriously affects patients themselves and their families seriously, but also places a heavy load on the society. Schizophrenia is a severe public health problem and a serious societal problem.

Schizophrenia is one of the most severe mental disorders and places a huge economic burden on the sufferer's family and on society. China is a developing country with the largest population in the world and expends a good deal of its health resources and services administration on the treatment and prevention of schizophrenia. Understanding the prevalence of schizophrenia in China has important implications for both health resources planning and risk factor epidemiology. Two large-scale psychiatric epidemiologic surveys conducted in 1982 and 1993 in China showed that the lifetime prevalence of schizophrenia was 5.69 per 1000 and 6.55 per 1000, respectively, while the point prevalence was 4.75 per 1000 and 5.31 per 1000 respectively (4, 5). Some provinces such as Hebei (6), Jiangxi (7) and Guangxi (8), and many cities have conducted epidemiological surveys on schizophrenia, but the results show great difference. The survey in Hebei showed that the lifetime and point prevalence of schizophrenia was 6.62 per 1000 and 5.46 per 1000 respectively (6); however, lifetime prevalence of schizophrenia of 2.88 per 1000 was estimated was estimated by Wang et al. (9) for a city in Jiangsu and the point prevalence of 7.96 per 1000 was found in a study by Deng et al. (10) in Sichuan. The prevalence of schizophrenia found was quite different, which may be related to latitude (11), or to differences between the diagnostic criteria, sample methods and investigators. In addition, different racial, cultural, educational and economic levels and genetic and biological factors may contribute (12, 13).

The latest national survey of mental illness was conducted in 1993. Since then, there has been no national epidemiological survey of mental illness. Mainland China has a vast territory, with the largest population in the world. Therefore, it is difficult to conduct an epidemiological survey on schizophrenia across the entire country. We were interested in exploring several research questions: What is the prevalence of schizophrenia in mainland China? Are estimates similar for males and females? Is there any urban–rural gradient? Would the estimates change over time? All those questions have been left unanswered, and there has not been a previous systematic review of epidemiological surveys on schizophrenia conducted in mainland China (not including Hong Kong, Taiwan and Macao).

Aims of the study

Based on the existing literature, we predicted that the pooled prevalence of schizophrenia in mainland China would be consistent with the international prevalence estimates. We predicted that there would be no significant difference in prevalence estimates between males and females, but that prevalence estimates would be higher in urban vs. rural regions.

Material and methods

Literature search

We conducted a systematic electronic search for published studies on the prevalence of schizophrenia (published from January 1980 to July 2013) in English databases such as 'PubMed' and 'Embase' and the four Chinese databases, which are the Biological Medical Literature database (CBM), Chinese National Knowledge Infrastructure (CNKI), Chongqing VIP and Chinese Wang Fang, using the key terms 'schizophrenia,' 'prevalence,' 'mental disorder,' 'mental illness,' 'epidemiology,' and 'China,'. We also searched the literature for references in each relevant study, or reviews. In relation to unpublished studies associated with the prevalence of schizophrenia, we were able to find them through bibliographic searches and send letters or e-mails to ask the authors for them.

Inclusion and exclusion criteria

The studies we finally selected were required to meet the following inclusion criteria in this study:

i) The literature was published between January 1980 and July 2013 in Chinese or English.
ii) The study was conducted wholly or partially in mainland China (except for Hong Kong, Taiwan and Macao).

iii) Studies were based on the general population sample but not special populations (prisoners, those in earthquake-stricken areas, the elderly, or volunteers).
iv) Studies provided the prevalence rather than the incidence of schizophrenia or provided other relevant information for the computation of prevalence for our study.
v) The territorial level of the study was above city level.

The exclusion criteria included the following:

i) Duplicated or repeated articles. For example, in the same year, a city-level article has been contained in another provincial-level article.
ii) Studies with unavailable data for our study or no survey date.
iii) Studies used census sampling as their method.
iv) Studies did not state the sample methods in the article.
v) Studies that corresponded to one of the three excluded criteria as below: i) Did not provide clear screening tools; ii) Screened only by trace survey, which only asked for the relevant information about the patients from people in the know without using effective screening tools in the screening period; iii) Did not use two-phase screening, in which the first phase was not carried out with screening tools like the Mental Illness/Health Screening Schedule (MISS/MHSS), Present State Examination (PSE), General Health Questionnaire (GHQ-12), Composite International Diagnostic Interview (CIDI) or individual health questionnaire, and the second phase was not screened by diagnostic criteria, such as those found in the Diagnostic and Statistical Manual of Mental Disorders, the Fourth vision (DSM-IV), International Classification of Diseases, 10th version (ICD-10) and International Classification of Diseases, 9th version (ICD-9).

Data extraction and analysis

We extracted the relevant data from the included studies to an Excel table. Data included author; year of publication; age range; survey date; sample methods; diagnostic criteria; diagnostic tools; screening tools; total sample size; lifetime prevalence and point prevalence of schizophrenia, including the overall prevalence; gender; urban or rural setting. The data extraction was finished by two authors independently. When disagreements on the results appeared, these were resolved together or by the third author.

Statistical analysis

With the help of Cochrane Review Manager 5.1 (The Cochrane Collaboration, The Nordic Cochrane Centre, Copenhagen, Denmark) and stata 11.1 (Stata Corporation, College Station, TX, USA) software, we conducted a meta-analysis to estimate the pooled prevalence of schizophrenia and analyze the prevalence by gender, location, year and province. The DeSimonian and Laird method (14) was used to calculate pooled estimates and 95% confidence intervals (95%CI) of schizophrenia prevalence. Odds ratio (OR) rate and 95% CI were used to compare differences between males and females and urban and rural (The urban/rural variables were defined according to their household registration. The system divided the residents into urban and rural according to their registration in the system no matter where they live. In the aspect of geographical distribution, usually people natively live in level regions above prefectural were urban household registration. In the aspect of geographical distribution, people natively live in level regions below township level were usually rural household registration). Heterogeneity between studies was detected by Q-test and I^2-statistics. The random-effects model was adopted when there was significant heterogeneity ($I^2 > 50\%$ or $P < 0.1$); otherwise, fixed-effects model was used.

Results

Study search

A total of 6902 studies published in PubMed ($n = 348$), Embase ($n = 103$), Chinese Wang Fang ($n = 986$), Chongqing VIP ($n = 493$), CNKI ($n = 4122$) and CBM ($n = 1198$) were identified, and 215 related studies were selected from all the databases by reading the titles and abstracts. After reading the full-text articles, as described in Supplementary material, we excluded 166 studies for the following reasons: i) Articles based on special population or surveys conducted in special areas ($n = 55$); ii) Articles repeated between provinces and cities, counties, and townships ($n = 42$); iii) Territorial level of the article survey site below the city level ($n = 51$, county level: 38, township level: 13); iv) No survey time ($n = 3$); v) Sample method of the study was census sampling ($n = 10$); vi) No sample method indicated ($n = 1$); vii) Studies did not provide clear screening tools or studies screened only by trace survey, or studies were not two-phase screening ($n = 4$). Finally, we had 49 studies suitable for this study (4–10, 15–56). The

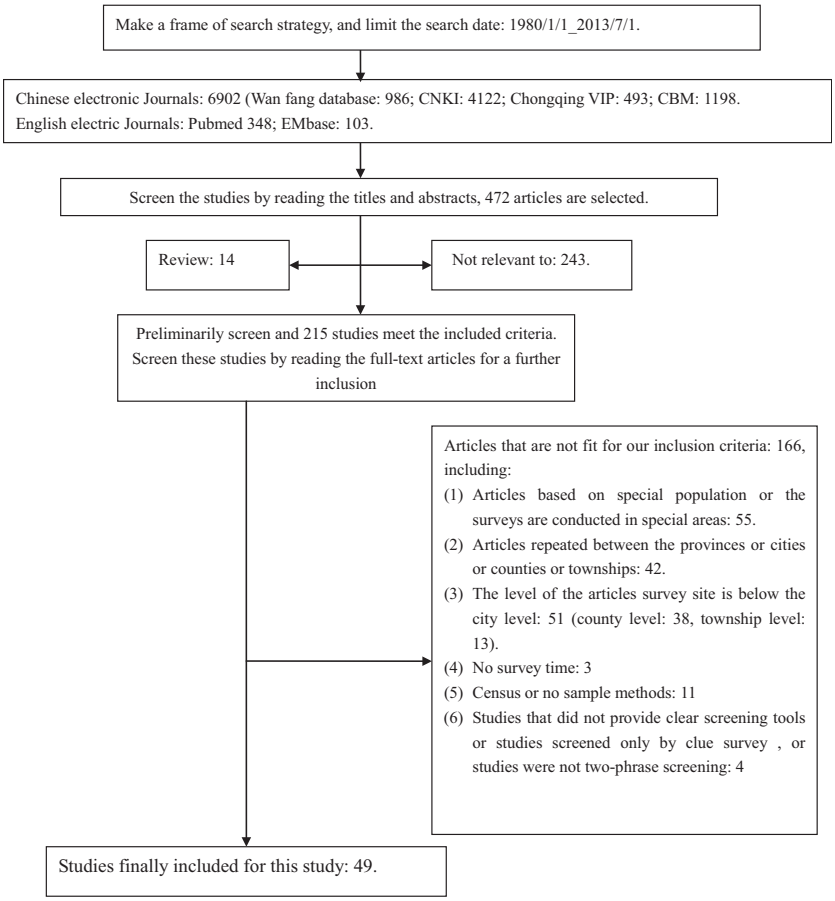

Fig. 1. The strategies of searching the literature.

procedure of the study research can be seen in Fig. 1.

Characteristic of the identified studies

We adopted 49 studies with a total sample size of 1 179 446 for this study, including two China-level (4, 5), 13 provincial-level (6–8, 15–24), and 34 city-level articles (9, 10, 25–56). These studies covered 19 provinces (Anhui: 2; Fujian: 2); Guangdong: 4; Gansu: 1; Guizhou: 1; Hainan: 1; Hebei: 1;, Heilongjiang: 2; Henan: 2; Hubei: 2; Jiangsu: 6; Jiangxi: 2; Jinlin: 1; Qinghai: 1; Shandong: 3; Shanxi: 1; Sichuan: 3; Yunnan: 3; Zhejiang: 4), one municipality (Beijing: 2), and three autonomous regions (Guangxi: 1; Xinjiang: 1; Tibet: 1). The lifetime prevalence of schizophrenia ranged from 2.88 per 1000 in Heilongjiang (21) to 9.77 per 1000 in Guangxi (8), and the point prevalence ranged between 1.79 per 1000 in Qinghai (17) and 8.29 per 1000 (in Guangxi) (8). A total of 39 of the studies we adopted were conducted based on the population aged 15 years old or older, four were based on those of 18 years old or older and six studies considered those of all ages. There are several kinds of diagnostic criteria commonly used in the studies, including: Chinese Classification and Diagnostic Criteria of Mental Disease, Second version (CCMD-2, n = 4), Chinese Classification of Mental Disorders, Second Edition-Revised (CCMD-2-R, n = 10), Chinese Classification and Diagnostic Criteria of Mental Disease, Third version (CCMD-3, n = 9), International Classification of Diseases, 9th version (ICD-9, n = 1), International Classification of Diseases, 10th version (ICD-10, n = 7), Diagnostic and Statistical Manual of Mental Disorders, Fourth version (DSM-IV, n = 11), and the diagnostic criteria in the Chinese Manual for Psychiatric Epidemiological Survey (CMPES, n = 11). All surveys of the included studies were conducted door-to-door using specific screening tools. Detailed information of study characteristics can be found in Table 1.

Prevalence of schizophrenia

Lifetime prevalence of schizophrenia. A total of 42 studies were identified for lifetime prevalence of schizophrenia. The overall lifetime prevalence was 5.44 per 1000 (95%CI: 5.01–5.88) (Table 2, Fig. 2). In terms of different provinces, as shown in Fig. 3, the highest lifetime prevalence—8.10

Table 1. Characteristic of the identified studies

Study	Survey date	Provinces	Territorial levels	(U&R)	Investigators	Sample methods	Screening/diagnostic tools/diagnostic criteria
Zhang et al. 1986 (4)	1982/7	China	China	U&R	Physicians; residents;nurses; statisticians	Multistage stratified cluster random	PSE/CMPES/CMPES
Weng et al. 1998 (22)	1984/7	Shandong	Provincial	U&R	Trained investigators	Stratified cluster random	PSE/CMPES/CMPES
Chen et al. 1988 (24)	1985/5	Fujian	Provincial	U&R	Psychiatrist	Urban: cluster sampling Rural: random sampling by economic stratification	PSE/CMPES/ICD-9
Duan et al. 1989 (56)	1985/10	Sichuan	City	U	Psychiatric professionals	Multistage cluster	PSE/CMPES/CMPES
Li et al. 1989 (55)	1986/3	Jilin	City	U&R	Trained investigators	Multistage stratified cluster	PSE/CMPES/CMPES
Hu et al. 1991 (53)	1905/6	Sichuan	City	U&R	Psychiatric specialist	Small group cluster random	CMPES/CMPES/CMPES
HangZhou team 1990 (54)	1988/6	Zhejiang	City	U&R	Epidemiological investigation group	Multiple cluster	PSE/CMPES/CMPES
Wang et al. 1995 (9)	1989/7	Jiangsu	City	U&R	–	Cluster random	CMPES/CMPES/CMPES
Li et al. 1994 (52)	1990/11	Heilongjiang	City	R	Psychiatrist	Cluster and system	MHSS/CMPES/CMPES
Guo et al. 1994 (51)	1991/8	Beijing	City	U&R	Investigation teams; psychiatric professionals	Stratified cluster random	CMPES/CMPES/CMPES
Tu et al. 1995 (50)	1992/11	Hubei	City	U&R	–	Two stage systematic	PSE/CMPES/CCMD-3
Chen et al. 1998 (5)	1993/4	China	China	U&R	Physicians; residents; nurses; statisticians	Multistage stratified cluster random	PSE/CMPES/CMPES
Ye et al. 1995 (23)	1994/4	Hainan	Provincial	U&R	Psychiatrist; trained investigators	Stratified random	CMPES/CMPES/CCMD-2
Hou et al. 1996 (49)	1994/12	Jiangsu	City	U&R	–	Urban: multiple cluster; rural: by economic stratification	PSE/CMPES/CCMD-2
Chang et al. 1997 (48)	1994/6	Jiangsu	City	U	Trained medical staffs	Random	CMPES/CMPES/CCMD-2
Weng et al. 1998 (22)	1994	Shandong	Provincial	U&R	Trained investigators	Cluster random	PSE/CMPES/CMPES
Shen et al. 1997 (47)	1995/3	Yunnan	City	U&R	Psychiatric professionals	Stratified cluster random	PSE/CMPES/CCMD-2
Zhang et al. 1997 (46)	1995/7	Guangdong	City	U&R	Attending physicians; physicians; nurses; psychological investigators	Random	PSE/CMPES/CCMD-2-R
Shi 1998 (45)	1995	Zhejiang	City	U&R	Trained backbones	Random	CMPES/CMPES/CCMD-2-R
Cheng et al. 1999 (44)	1996/7	Guangdong	City	U&R	Trained investigators	Stratified cluster	PSE/CMPES/CCMD-2-R
Liu et al. 2000 (21)	1997/4	Heilongjiang	Provincial	U&R	Trained investigators	Stratified cluster	PSE/CMPES/CCMD-2-R
Cheng et al. 2001 (43)	1998/5	Hubei	City	U&R	Psychiatrist	Random table cluster	CMPES/CMPES/CCMD-2-R
Li et al. 2003 (39)	1998/8	Jiangsu	City	U&R	Psychiatrist	Cluster multiple random	PSE/CCMD-2-R/CCMD-2-R
Wan et al. 2002 (42)	1999/7/1	Jiangxi	City	U&R	Physicians	Stratified cluster random	PSE/CMPES/CCMD-2-R
Deng et al. 2003 (10)	2000/11	Sichuan	City	U&R	Psychiatrist	Multistage cluster	PSE/CMPES/CCMD-2-R
Wang et al. 2002 (41)	2000/11	Anhui	City	U&R	–	Stratified cluster random	PSE/CMPES/CCMD-2-R
Hu et al. 2003 (40)	2000/11	Guangdong	City	U&R	Trained investigators	Stratified cluster random	MHSS/CMPES/CCMD-2-R
Zhang et al. 2003 (20)	2001/7	Guizhou	Provincial	U&R	Psychiatrist	Cluster	PSE/CMPES/CCMD-3
Shi et al. 2005 (18)	2001/9	Zhejiang	Provincial	U&R	Psychiatric specialist; psychiatric nurses	Multistage stratified cluster	GHQ-12/SCID-I/P/DSM-IV
Lu et al. 2004 (7)	2002/3	Jiangxi	Provincial	U&R	Psychiatrist	Primary unit content proportional stratified cluster and random	CIDI/CIDI/ICD-10
Tang et al. 2005 (36)	2002/12	Zhejiang	City	U&R	–	Primary unit content of stratified cluster random	MISS/CIDI/ICD-10
You et al. 2003 (38)	2003/3	Henan	City	U&R	–	Primary unit content of stratified cluster random	CIDI-C/CMPES/CCMD-3
Wei et al. 2004 (19)	2003/7	Tibet	Provincial	U&R	Huaxi institute of mental health professionals; non-psychiatric medical staffs	Classification quota random cluster	MHSS/SCID-I/P-I/DSM-IV
Yuan et al. 2004 (37)	2003/4	Jiangsu	City	U&R	Psychiatric medical staffs	Multistage cluster random	PSE/CMPES/CCMD-3
Xiang et al. 2008 (33)	2003/4	Beijing	City	U&R	–	Stratified multistage systematic selection	CIDI-1.0/CIDI-1.0/ICD-10
Cui et al. 2007 (6)	2004/10	Hebei	Provincial	U&R	Psychiatric specialist; psychiatric nurses	Multistage stratified cluster random	GHQ-12/SCID-I/P/DSM-IV
Zhang et al. 2012 (15)	2004/12	Shandong	Provincial	U&R	Psychiatrist; psychiatric nurses	Multistage stratified cluster random	GHQ-12/SCID-I/P/DSM-IV
Shen et al. 2007 (35)	2005/3	Jiangsu	City	U&R	Trained investigators; psychiatrist	Stratified cluster random	PSE/CMPES/CCMD-3
Ding et al. 2010 (30)	2005/4	Gansu	City	U&R	Psychiatrist; nurses	Multistage stratified cluster random	GHQ-12/SCID-I/P/DSM-IV

Table 1. (Continued)

Study	Survey date	Provinces	Territorial levels	(U&R)	Investigators	Sample methods	Screening/diagnostic tools/diagnostic criteria
Ruan et al. 2010 (27)	2005/11	Yunnan	City	U&R	Psychiatric medical staffs; medical graduates	Stratified three-stage capacity proportion probability random	CIDI-2.1/CIDI-2.1/DSM-IV/ICD-10
Song et al. 2010 (17)	2005/3	Qinghai	Provincial	U&R	Psychiatrist; psychiatric nurses	Multistage stratified cluster random	GHQ-12/SCID-I/P/DSM-IV
Dong et al. 2008 (34)	2006/9	Shandong	City	U&R	–	Stratified cluster random	PSE/CMPES/CCMD-3
Zhao et al. 2009 (29)	2006/8	Guangdong	City	U&R	Trained investigators; psychiatrist	Stratified cluster random	CIDI-3.0/CIDI-3.0/DSM-IV
Zhan et al. 2011 (26)	2006/8	Yunnan	City	U&R	Psychiatrist	Multistage stratified cluster random	CMPES/CMPES/CIDI/CCMD-3/ICD-10
Wei et al. 2011 (8)	2007/7	Guangxi	Provincial	U&R	Psychiatry professionals; psychiatrist	Multistage stratified cluster random	CIDI-3.0/CIDI-3.0/ICD-10
Xu et al. 2009 (32)	2007/1	Shanxi	City	U&R	Psychiatric professionals	Multistage stratified cluster random	MISS/CMPES/CIDI/CCMD-3/ICD-10
Yao et al. 2009 (31)	2007/3	Anhui	City	R	Trained investigators; psychiatrist	Random	PSE/CMPES/CCMD-3
Gao 2010 (28)	2008/10	Xinjiang	City	U&R	Doctors; nurses	Multistage cluster random	GHQ-12/SCID-I/P/DSM-IV
Fang et al. 2011 (16)	2009/2	Fujian	Provincial	U&R	Psychiatric specialist; psychiatric nurses	Multistage stratified cluster random	GHQ-12/SCID-I/P/DSM-IV
Liu et al. 2012 (25)	2011/8	Henan	City	U&R	All levels of medical staffs; Mental health professionals; psychiatrist	Random	Individual health questionnaire/SCID-I/P/DSM-IV

U&R, Urban & Rural; HangZhou team, Hangzhou epidemiological investigation team of the seventh people's hospital; PSE, Present State Examination; MISS, Mental Illness Screening Schedule; MHSS, Mental Health Screening Schedule; GHQ-12, General Health Questionnaire; CIDI, Composite International Diagnostic Interview; SCID-I/P, Structured Clinical Interview for DSM-IV-TR Axis I Disorders-Patient Edition; CCMD-2, Chinese Classification and diagnostic criteria of Mental Disease, the Second version; CCMD-2-R, Chinese Classification of Mental Disorders, Second Edition, Revised; CCMD-3, Chinese Classification and diagnostic criteria of Mental Disease, the Third version; ICD-9, International Classification of Diseases, The 9th version; ICD-10, International Classification of Diseases, The 10th version; DSM-IV, Diagnostic and Statistical Manual of Mental Disorders, the Fourth version; CMPES, diagnostic criteria in the Chinese Manual for Psychiatric Epidemiological Survey.

per 1000–9.77 per 1000—was found in Guangxi, Shanxi, and Gansu.

Lifetime prevalence by gender. Only 25 studies were available for lifetime prevalence separated by gender. The lifetime prevalence of males and females is 5.33 per 1000 (95%CI: 4.55–6.11) and 5.51 per 1000 (95%CI: 4.75–6.26) (Table 2, Fig. 2) respectively. No statistically significant difference was found between males and females; the OR of males vs. (VS) females was 0.97 (95% CI: 0.81–1.17).

Lifetime prevalence by location. A total of 26 studies were used to estimate the lifetime prevalence calculation when separated by location. The lifetime prevalence of urban and rural dwellers was 6.60 per 1000 (95%CI: 5.83–7.36) and 4.73 per 1000 (95%CI: 4.16–5.30) (Table 2) respectively. The OR of schizophrenia in urban and rural areas was 1.44 (95%CI: 1.30–1.59), indicating that people in urban areas were 1.44 times more likely to have schizophrenia than those in rural areas.

Lifetime prevalence by years. A total of 42 studies were identified to be related with the lifetime prevalence of schizophrenia from 1980 to 2013. When analyzing by years, prevalence ranged from 2.85 per 1000 to 8.41 per 1000. When analyzing by periods, the lifetime prevalence of schizophrenia in the periods of 1980–1990, 1990–2000 and 2000–2013 were 4.71 per 1000, 5.61 per 1000, and 5.96 per 1000 respectively (Table 2).

Point prevalence of schizophrenia

A total of 36 studies were available for point prevalence. As shown in Table 2 and Fig. 4, the point prevalence of schizophrenia is 4.62 per 1000 (95% CI: 4.18, 5.06). When analyzing by provinces, we found that the prevalence in Guangxi (8.29 per 1000) was highest (Fig. 5).

Point prevalence by gender. A total of 18 studies provided us with the point prevalence of schizophrenia for males and females. The point prevalence of males and females is 4.63 per 1000 (95% CI: 3.93–5.34) and 4.95 per 1000 (95%CI: 4.21–5.70) (Table 2, Fig. 4) respectively. There was no significant difference between males and females; the OR of males and females was 0.91 (95%CI: 0.79–1.05).

Point prevalence by location. A total of 22 studies were used in this study for the point prevalence of

Table 2. The lifetime and point prevalence of schizophrenia

Items	Amount of related articles		Total sample size		SCZ cases		Prevalence (per 1000); [95%CI]		I^2 (%)	
	LP	CP	LP	CP	LP	CP	LP	CP	LP	CP
Overall	42	36	1 039 822	722 755	4947	3144	5.44 [5.01–5.88]	4.62 [4.18–5.06]	91.5	86.6
Gender										
Male	25	18	346 104	201 107	1584	842	5.33 [4.55–6.11]	4.63 [3.93–5.34]	90.2	81.2
Female	25	18	346 840	204 500	1574	965	5.51 [4.75–6.26]	4.95 [4.21–5.70]	89.0	82.0
Location										
Urban	26	22	357 481	174 013	2053	952	6.60 [5.83–7.36]	5.15 [4.40–5.89]	83.3	77.8
Rural	26	22	504 189	327 043	1992	1307	4.73 [4.16–5.30]	4.44 [3.84–5.05]	86.7	84.6
Years										
[1980–1990)	8	6	517 410	165 145	2063	625	4.71 [3.81–5.61]	4.34 [3.32–5.36]	94.7	85.1
[1990–2000)	15	9	238 297	174 809	1230	806	5.61 [4.69–6.52]	4.96 [4.01–5.92]	87.2	84.5
[2000–2013]	20	22	284 115	382 801	1654	1713	5.96 [5.15–6.78]	4.59 [3.97–5.21]	87.6	87.7

SCZ, schizophrenia; LP, Lifetime prevalence; CP, Current prevalence; CI, Confidence Interval.

schizophrenia. The urban v. rural prevalence was 4.65 per 1000 (95%CI: 4.17–5.12) v. 4.44 per 1000 (95%CI: 3.84–5.05) (Table 2). The OR (OR = 1.20, 95%CI: 1.02–1.41) of point prevalence in urban and rural areas indicated that people in urban areas were 1.20 times more likely to have schizophrenia than people in rural areas.

Point Prevalence by years. A total of 36 studies were useful for the point prevalence of schizophrenia from 1980–2013. When analyzing point prevalence by years, the prevalence varied from 2.56 per 1000 to 7.83 per 1000. When analyzing by periods, the point prevalence of schizophrenia in the periods of 1980–1990, 1990–2000 and 2000–2013 were 4.34 per 1000, 4.96 per 1000, and 4.59 per 1000 respectively (Table 2).

Discussion

This is the first meta-analysis focusing on the prevalence of schizophrenia in mainland China since the two national epidemiological surveys on schizophrenia conducted in 1982 and 1993. In keeping with these two national studies, in 1982 (lifetime: 5.69 per 1000, point: 4.75 per 1000) (4) and 1993 (lifetime: 6.55 per 1000, point: 5.31 per 1000) (5), our results showed that the lifetime and point prevalence of schizophrenia was 5.44 per 1000 and 4.62 per 1000, respectively, which indicated that the prevalence in mainland China remained at a stable level. When analyzing by gender, there was no evidence that showed a significant difference between males and females, however, when analyzing by location it was found that people in urban areas face a higher risk of schizophrenia than those in rural areas.

The prevalence rate of schizophrenia in mainland China was at a relatively high prevalence rate. Some studies estimated lower prevalence of schizophrenia than that of in mainland China. A systematic review of schizophrenia by Saha et al. (1) in 2005 found that the median prevalence of schizophrenia in developing counties (including China) was 2.6 per 1000, while in developed countries, it was 3.3 per 1000. And an epidemiology study of schizophrenia showed that the prevalence of schizophrenia had a range of 1.4 per 1000~4.6 per 1000 (57). A similar low prevalence rate was found in a systematic review of prevalence of schizophrenia across the population in 2002, which found that the pooled prevalence in Asian (Hong Kong, Taiwan, and Korea) was 2.5 per 1000 (95% CI: 0.19–0.32) (58). In India, surveys on schizophrenia in 1998 and 2000 reported the prevalence of schizophrenia was 2.7 per 1000 and 2.5 per 1000 respectively (59). Similarly, a survey on schizophrenia in Nithsdale, South West Scotland, estimated the point prevalence was 3.19 per 1000 (60). Several reasons may explain the relatively high prevalence rate of schizophrenia in mainland China. First, rapid development and a large population contribute to fiercer competition in mainland China, which may lead to a higher risk of schizophrenia. According to the Sixth National Population Census in 2010, there were at that time about 1.34 billion people in mainland China (61). And along with the expansion of economic opportunity, work-related stress appears to be increasing due to increased competition (62). Second, great differences in social geography, culture, and customs, lower-level economics, and less-advanced working patterns in mainland China should be taken into account (1). Third, the cross-cultural reliability and validity of some instruments in Eng-

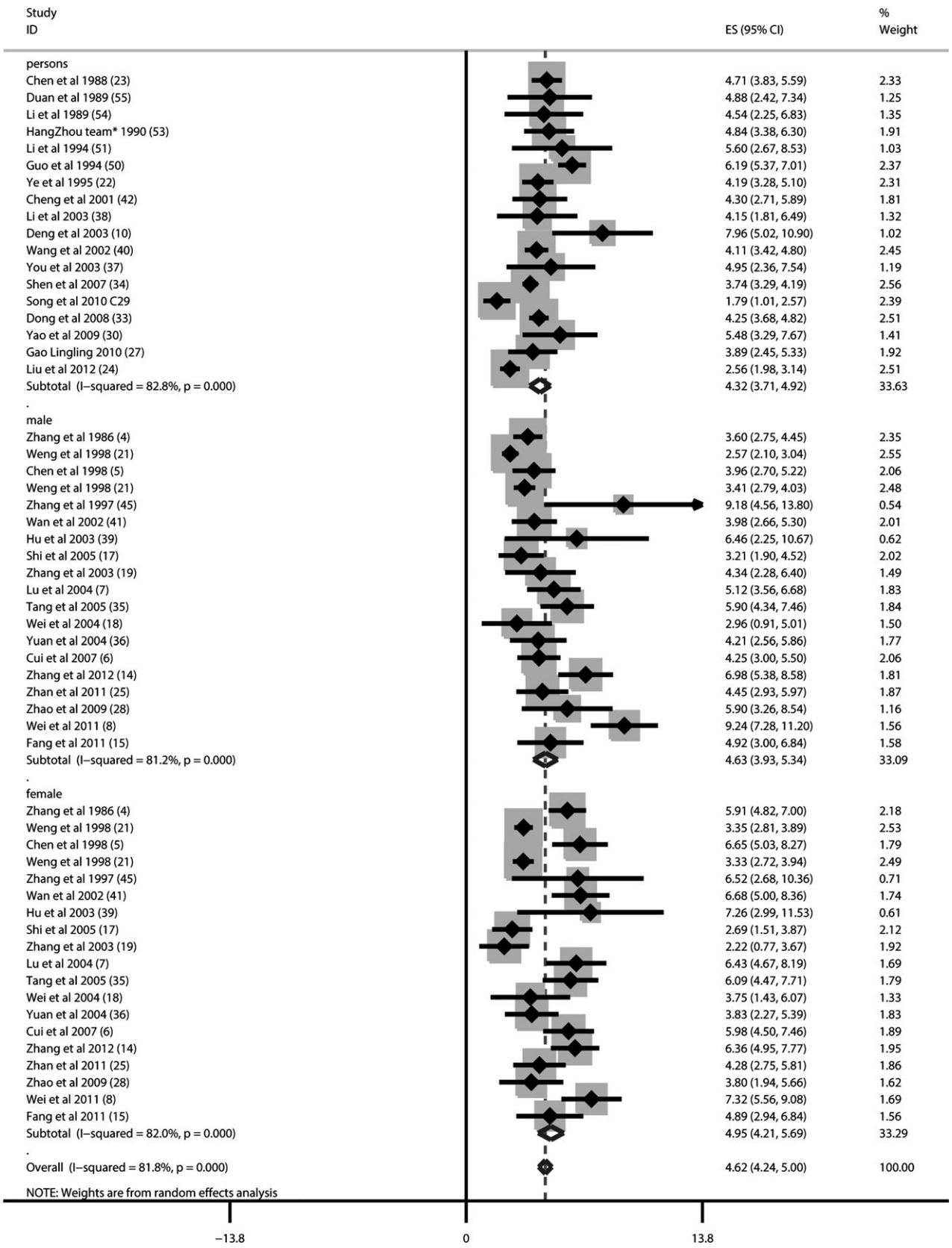

Fig. 2. The lifetime prevalence of schizophrenia and analyzing by gender.

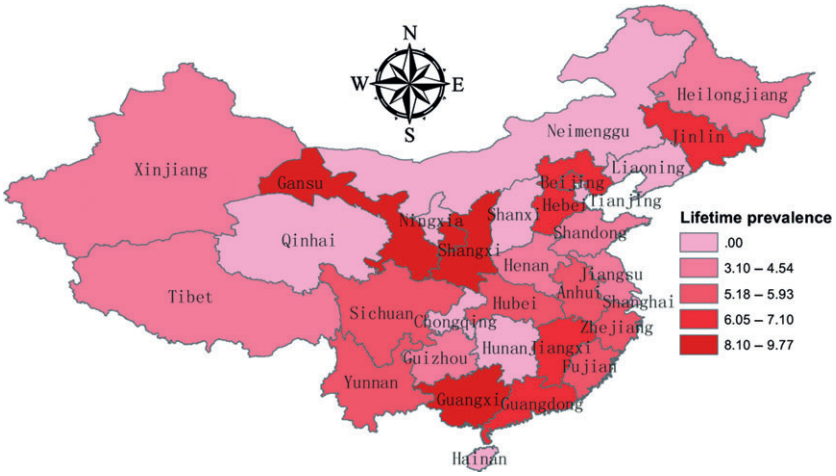

Fig. 3. Estimate the lifetime prevalence of schizophrenia by provinces.

lish, such as CIDI, remain unclear because the validation exercises have been completed almost entirely in Western countries. Thus, the diagnostic tools and screening tools also played an indispensable role in the variation of prevalence rate (63). Fourth, it may be related to less awareness of schizophrenia in mainland China (64). However, some studies found higher prevalence of schizophrenia than that of in mainland China. A study in Kosrae, Micronesia, found the point prevalence in the population of 15 years old or older was 6.8 per 1000 (65), and a study on the prevalence of depression among households in three major cities in Pakistan referred to a pooled prevalence of schizophrenia of 15 per 1000 (66). In addition, in a study from Finland on general population, the lifetime prevalence of schizophrenia was 8.7 per 1000 (67).

Prevalence by location

Higher prevalence of schizophrenia was found in urban areas in this systematic review of studies in mainland China. Similar results can be found in other studies. A meta-analysis of the association of urbanicity with schizophrenia showed that schizophrenia risk in the most urban areas was estimated to be 2.37 times higher than in the most rural areas (68). Another meta-analysis on schizophrenia and urbanicity indicated that the prevalence rate of schizophrenia was around double the rate of that in rural areas (69). However, a systematic review in 2005 found that there was no significant difference between the median prevalence in urban (2.9 per 1000) and rural (2.95 per 1000) settings (1). The difference between urban and rural settings may be explained as follows. People in urban areas undertake fast-moving and stressful work patterns and experience much more psychological pressure, and more opportunities to gain a higher educational level and upward social mobility will contribute to mental health problems. Even the dietary structure could have an effect (70–72). In rural areas, meanwhile, less awareness of schizophrenia, lower educational attainment, higher rate of unemployment, backward medical conditions, lower economic status, less access to health services, and lower income contributes to the possibility that people in rural areas may conceal their condition even if they have schizophrenia (73, 74).

Prevalence by gender

We found no statistical difference between males and females in our systematic review, which was congruent with some previous studies. Okkels et al. estimated the incidence of early onset schizophrenia in Denmark using a nationwide, population-based, mental health register. Their findings suggest that the sex differential has narrowed over time and the usual male excess has disappeared in recent years (75). A comprehensive literature review focusing on gender differences in schizophrenia showed that no epidemiological studies have found a significant difference in males and females (76). In a concise overview of incidence, prevalence, and mortality of schizophrenia, no significant gender difference could be found; the median lifetime prevalence of males and females was 3.70 per 1000 and 3.80 per 1000 and the median point prevalence was 4.30 per 1000 and 3.00 per 1000 respectively (77).

Limitations

Several potential limitations have to be taken into consideration in our systematic review. First, a lack of data on the prevalence of schizophrenia could make a difference to the results. We identified 49 studies that covered 19 provinces, three autonomous regions (Guangxi, Xinjiang, Tibet),

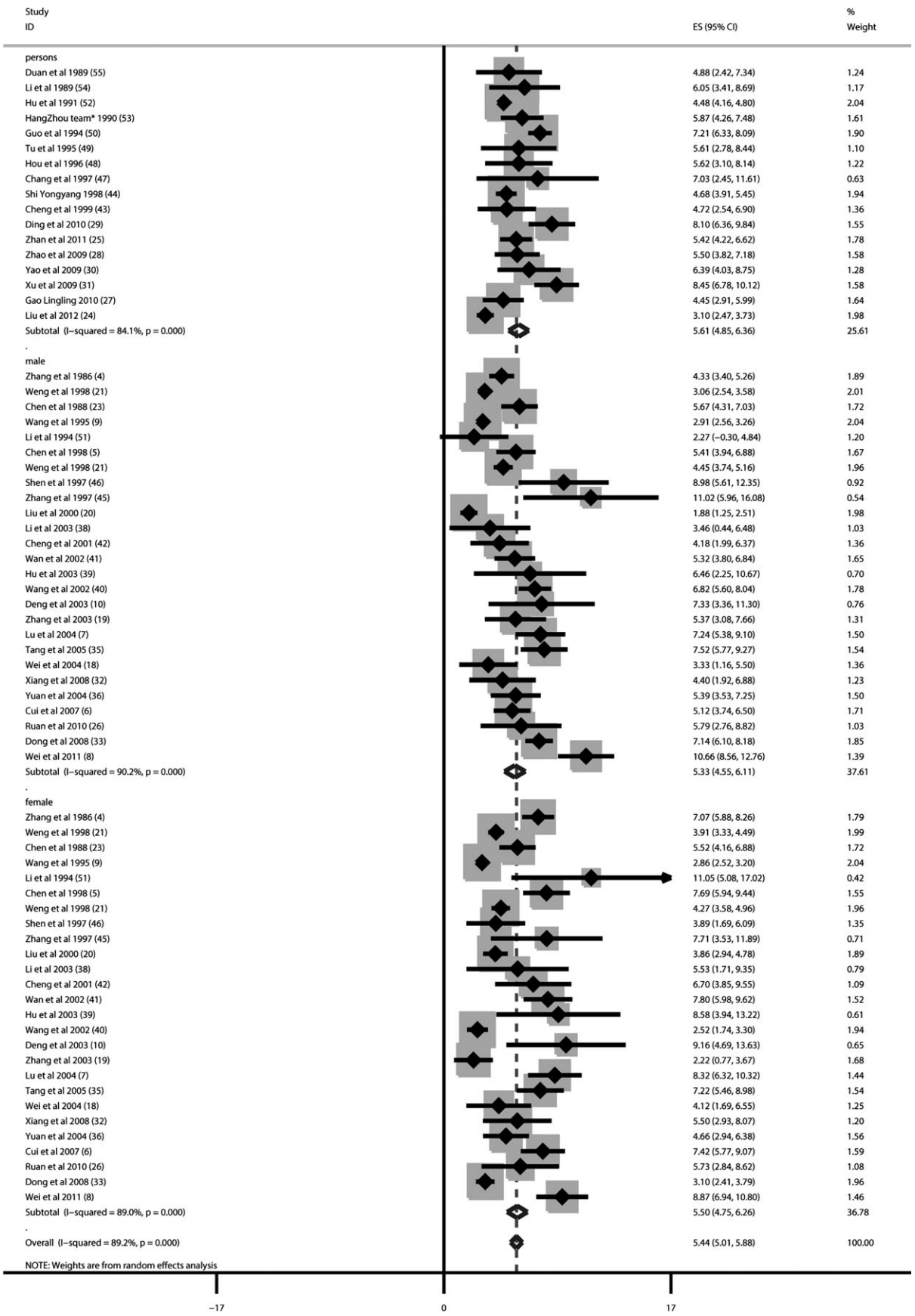

Fig. 4. The point prevalence of schizophrenia and analyzing by gender.

Fig. 5. Estimate the point prevalence of schizophrenia by provinces.

and one municipality (Beijing) from mainland China for use in our study. Chongqing, Tianjin, Shaanxi Province, Nei Monggol Autonomous Region, and Ningxia Hui Autonomous Region have not conducted psychiatric epidemiologic studies on schizophrenia. Moreover, not all of the 49 studies we identified as being available for the systematic review provided all the data we needed for prevalence estimate. Only several studies provided the completed data for schizophrenia (lifetime prevalence: 42 and point prevalence: 36). Second, unequal heterogeneity existed within each study and could not be adjusted, leading to limitations. Heterogeneity cannot be avoided when conducting a meta-analysis review (78), especially meta-analysis based on epidemiological surveys (79); this is caused by the different methodologies. With respect to our study, several methodological matters have an effect on the potential bias of the heterogeneity in the prevalence: the gender factor, the urban/rural factor, years, and different regions factors. All contribute to the heterogeneity of the estimated prevalence. Thus, in this systematic review estimating the prevalence of schizophrenia, we would like to draw attention to not only the results of schizophrenia but also the analysis of the results by subgroup. Third, as time went by, the screening tools, diagnostic criteria, and other methodological tools changed. In addition, some of the diagnostic tools were originally written in English and only tested in the native population before being translated to Chinese and used with local people in mainland China; thus, this limitation cannot be avoided. A final limitation relates to geographical differences. These include cross-cultural issues, different customs, and the varying policies of different local governments.

In conclusion, the results showed the lifetime and point prevalence of schizophrenia in mainland China is 5.44 per 1000 and 4.62 per 1000 respectively. There was no statistically significant difference found between males and females. However, a higher prevalence in urban areas than in rural dwellings was estimated in our study. Considering some provinces in mainland China, such as Shaanxi Province, have not previously conducted an epidemiological survey on schizophrenia, it is necessary to conduct more epidemiological surveys throughout mainland China.

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Declaration of interest

All authors declare no conflict of interests.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Data S1. Supplementary material.